THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 09

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For $\sqrt{12}$, Student A says $2\sqrt{3}$ (irrational) and Student B says 6 (rational). Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

2. A student states: "$\sqrt{9}$ is irrational because it has a root sign." The correct response is: 1

(a) Correct **(b)** Incorrect; $\sqrt{9} = 3$ is rational

(c) Correct for odd numbers **(d)** Cannot be determined

3. A quadratic polynomial whose zeroes are 4 and -1 is: 1

(a) $x^2 + 3x - 4$ **(b)** $x^2 - 3x - 4$ **(c)** $x^2 - 5x + 4$ **(d)** $x^2 + 3x + 4$

4. The graph of $y = ax^2 + bx + c$ with $a < 0$ is a parabola that opens: 1

(a) downwards **(b)** to the left **(c)** to the right **(d)** upwards

5. The solution of $2x + y = 7$ and $x - y = 2$ is: 1

(a) $(4, -1)$ **(b)** $(3, 1)$ **(c)** $(1, 3)$ **(d)** $(2, 3)$

6. The sum of two numbers is 20 and their difference is 4. The numbers are: 1

(a) $14, 6$ **(b)** $16, 4$ **(c)** $12, 8$ **(d)** $10, 10$

7. The roots of $3x^2 - 27 = 0$ are: 1

(a) $\pm\sqrt{3}$ **(b)** ± 3 **(c)** 3 only **(d)** ± 9

8. If $x = -\frac{1}{2}$ is a root of $3x^2 + 2kx - 3 = 0$, then k is: 1

(a) $-\frac{4}{9}$ **(b)** $\frac{9}{4}$ **(c)** 2 **(d)** $-\frac{9}{4}$

9. The next term of the AP $5, 2, -1, \ldots$ is: 1

(a) −2 **(b)** −3 **(c)** 4 **(d)** −4

10. If $a_n = 3 + 4n$, the common difference is: 1

(a) 4 **(b)** 3 **(c)** 7 **(d)** 1

11. A 6 m pole casts a 4 m shadow. At the same time a building casts an 18 m shadow. The height of the building is: 1

(a) 12 m **(b)** 27 m **(c)** 30 m **(d)** 24 m

12. A student states: "Two triangles are similar if two sides of one are proportional to two sides of the other." The correct response is: 1

(a) Correct
(b) Incorrect — the included angles must also be equal (SAS)
(c) Correct only for right triangles
(d) Cannot be determined

13. $(\sin 30^\circ + \cos 30^\circ) - (\sin 60^\circ + \cos 60^\circ)$ equals: 1

(a) 0 **(b)** 1 **(c)** $\sqrt{3}$ **(d)** −1

14. PA and PB are tangents to a circle of radius 6 cm with $\angle APB = 90^\circ$. Then OP is: 1

(a) 12 cm **(b)** $6\sqrt{2}$ cm **(c)** $3\sqrt{2}$ cm **(d)** 6 cm

15. A student states: "The length of a semicircular arc is πr^2." The correct response is: 1

(a) Correct **(b)** Incorrect — it is πr
(c) It is $2\pi r$ **(d)** Cannot be determined

16. If the area of a sector of a circle of radius 6 cm is 6π cm², its angle is: 1

(a) 90° **(b)** 45° **(c)** 60° **(d)** 30°

17. The curved surface area of a cylinder of radius 7 cm and height 10 cm is: 1

(a) 220 cm² **(b)** 1540 cm² **(c)** 308 cm² **(d)** 440 cm²

18. The class mark of the class 10–20 is: 1

(a) 30 **(b)** 15 **(c)** 20 **(d)** 10

19. In $\bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i}$, d_i is: 1

(a) $x_i + a$ **(b)** $x_i - a$ **(c)** $\frac{x_i - a}{h}$ **(d)** $a - x_i$

20. A letter is chosen from the word PROBABILITY. The probability that it is B is: 1

(a) $\frac{2}{11}$ **(b)** $\frac{2}{10}$ **(c)** $\frac{3}{11}$ **(d)** $\frac{1}{11}$

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Explain why $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ is a composite number. 2

22. Find the zeroes of $4s^2 - 4s + 1$ and verify the relationship between the zeroes and the coefficients. 2

23. Find the discriminant of $3x^2 - 4\sqrt{3}x + 4 = 0$ and hence the nature of its roots. Find the roots if real. 2

24. Give two different examples each of 2
(i) similar figures
(ii) figures that are not similar.

25. Find the point on the x-axis which is equidistant from $(2, -5)$ and $(-2, 9)$. 2

26. Evaluate: $\frac{\sin 30^\circ + \tan 45^\circ - \csc 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$. 2

27. A circus artist climbs a 20 m long rope tightly stretched and tied from the top of a vertical pole to the ground. Find the height of the pole if the rope makes 30° with the ground. 2

28. Prove that the perpendicular at the point of contact to the tangent of a circle passes through the centre. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $x + y = 6$ and $x - y = 2$. **3**

30. Find the mode of: **3**

Class	0–10	10–20	20–30	30–40	40–50
Frequency	2	5	9	7	3

31. Find the 12th term from the end of the AP $3, 5, 7, \ldots, 201$. **3**

32. Prove that the line joining an external point to the centre of a circle bisects the angle between the two tangents drawn from that point. **3**

33. Find the area of the minor segment of a circle of radius 20 cm if the angle of the sector is $60°$. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) **3**

34. All four aces are removed from a deck. One card is drawn from the rest. Find P(king), P(red card) and P(ace). **3**

35. The angle of elevation of the top of a hill from the foot of a tower is $60°$ and the angle of elevation of the top of the tower from the foot of the hill is $30°$. If the tower is 50 m high, find the height of the hill. **3**

OR

From a point on the ground 40 m away from the foot of a tower, the angle of elevation of the top of the tower is $30°$. The angle of elevation of the top of a water tank on the tower is $45°$. Find **3**
(i) the height of the tower
(ii) the depth of the tank.

36. The mean of the following data is 50. Find p. **3**

Class	10–30	30–50	50–70	70–90
Frequency	5	8	p	7

OR

Find the median from the cumulative frequency table: **3**

Marks less than	10	20	30	40	50	60
Students	5	12	25	38	45	50

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** On a town map drawn on a grid, a school is at S(2, 3) and a hospital at H(10, 9). **5**
(i) Find the distance SH.
(ii) A park P is to be built on SH dividing it in the ratio 1 : 3. Find P.
(iii) A market M is at the mid-point of SH. Find M.
(iv) Verify that M is equidistant from S and H.

OR

The vertices of $\triangle ABC$ are A(4, 6), B(1, 5) and C(7, 2). **5**
(i) Find the mid-points D of AB and E of AC.
(ii) Show that $DE = \frac{1}{2}BC$.
(iii) What does this verify?

38. **Case study:** A carpenter's set-square is a right triangle PQR, right-angled at Q, with $PQ = 8$ cm and $QR = 6$ cm. 5

(i) Find PR.

(ii) Find $\sin P$ and $\cos R$.

(iii) Why is $\sin P = \cos R$?

(iv) Show that $\tan P \cdot \tan R = 1$.

OR

Prove that $\sqrt{\frac{\sec\theta - 1}{\sec\theta + 1}} + \sqrt{\frac{\sec\theta + 1}{\sec\theta - 1}} = 2\csc\theta$. 5

39. **Case study:** A toy is a cone of base radius 7 cm and height 24 cm mounted on a hemisphere of the same radius. 5

(i) Find the slant height.

(ii) Find the surface area of the toy.

(iii) Find the cost of painting it at ₹0.50 per cm².

(iv) Find the volume of the toy.

OR

Case study: A pencil is a cylinder of length 15 cm and radius 0.35 cm with a conical tip of height 1.2 cm. 5

(i) Find the volume of the cylindrical part.

(ii) Find the volume of the tip.

(iii) Find the total volume.

(iv) Find the slant height and curved surface area of the tip.

40. **Case study:** Monthly electricity bills of 50 houses are given. 5

(i) Write the modal class.

(ii) Find the mode.

(iii) Find the mean by the assumed mean method.

Bill (₹)	400–600	600–800	800–1000	1000–1200	1200–1400
Houses	7	10	15	12	6

OR

A student obtains the median of the data below as 35. 5

(i) Find the correct median.

(ii) Find the mean by the step-deviation method.

(iii) What error could the student have made?

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	10	18	12	5

END OF QUESTION PAPER **Total: 80 Marks**